🎓 CAPITAL UNIVERSITY

Alumni Portal

A Bridge Between the University and its Graduates

Faculty of Computers & Artificial Intelligence | Software Engineering Department

Meet The Team

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CHAPTER 1

Introduction

Problem Statement, Proposed Solution, Objectives, & Stakeholders

Current Alumni Challenges

⚠ The Core Problem: Existing traditional methodologies introduce critical connectivity barriers.

Lost Connection

Graduates lose active contact with the university instantly after graduation, creating a lifetime data gap.

No Network

Absence of a professional centralized platform for alumni networking and continuous engagement.

Manual Services

Document requests, official transcripts, and degree verification are handled through outdated, manual processes.

The Proposed Solution

Centralized Web Platform

An innovative, highly secure web application designed to orchestrate three main actors:

- **Alumni:** Effortless community networking and services.
- **University Staff:** Efficient process automators.
- **Super Admin:** Total governance and configuration.

Rich Feature Ecosystem

More than just a social platform; it offers an integrated administrative service pipeline:

- **Networking:** Professional profiles, communities, and live chat.
- **Services:** Secure digitized Document Request System.
- **Opportunities:** Career board and announcements channel.

Key Project Objectives

- **Strengthen Bonds:** Build a persistent, lifelong relationship channel between graduates and their Alma Mater.
- **Professional Ecosystem:** Establish a trusted directory to help alumni easily track peer career trajectories.
- **Modernize Workflows:** Fully digitize time-consuming services like Document Requesting & Verification.
- **Real-Time Delivery:** Enable fast message propagation and instant status updates via Sockets.
- **Robust Governance:** Enforce high security with state-of-the-art RBAC architecture.

Stakeholder Responsibilities (RBAC)

System Governance: Granular access control mapping user operations.

Alumni Role

Manage career profile, request verified documents, post updates, join specialized communities, and engage in live peer chat.

Staff Role

Review and process incoming document requests, publish critical academic announcements, and moderate university communities.

Super Admin

Oversee total system operation, manage staff and alumni user records, configure granular permission tables, and audit security logs.

CHAPTER 2

Requirements Analysis

Functional Requirements, Non-Functional Requirements, & User Roles

Alumni Key Features

Verification & Professional Tools

- **Digital ID + QR Code:** instant official alumni verification.
- **Document Request System:** Fully digital request workflow.
- **Friend Recommendation System:** Network matching algorithm.
- **Profile Management:** Detailed portfolio representation.

Engagement & Communication

- **Real-Time Chat:** Direct messaging backed by Socket.io.
- **Alumni Communities:** High-interaction groups by majors.
- **Job Opportunities:** Tailored professional job board.
- **Real-time Notifications:** Push alerts on status changes.

CHAPTER 3

System Design

Architecture, Database Design, UML Modeling, & Workflows

System Tech Stack & Architecture

Frontend

React.js

SPA architecture delivering responsive user experience and component-based structure.

Backend

Node.js / Express.js

Asynchronous, highly scalable REST API layer handling server logic and data routing.

Database

PostgreSQL / Sequelize

Relational database mapping out robust schemas with strict integrity rules through an ORM.

Real-Time

Socket.io / Cloudinary

Low latency real-time communication coupled with CDN media distribution.

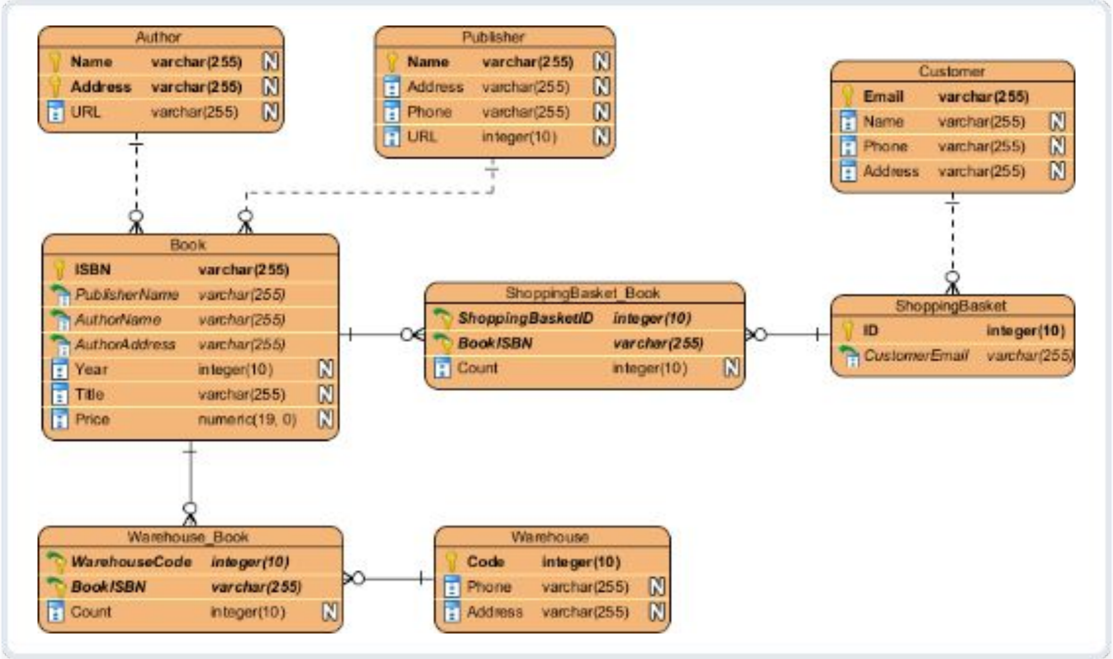
System Design Models

ARCHITECTURAL BLUEPRINTS

Standardized Engineering

Our modeling methodology follows standardized industry rules to guarantee future-proof code maintainability:

- **Class Diagram:** Maps OOP object relationships.
- **Entity Relationship Diagram (ERD):** Relational schema structuring DB data.
- **Use Case Diagram:** Defining roles actions.
- **Sequence Diagrams:** Dynamic API message flow.
- **Activity Diagrams:** Step-by-step business workflows.



Document Request System Workflow

Automated Verification Pipeline: A secured systematic transition state model.

1

Submission

Alumni initiates the request via the portal dashboard.

2

Upload

Alumni uploads essential verification files securely.

3

Persistence

Data commits to PostgreSQL with a 'Pending' status flag.

4

Notification

Socket-based alert triggers on the admin/staff panel.

5

Audit

Staff reviews document authenticity carefully.

6

Update & Track

Status shifts to Approved/Rejected; Alumni notified instantly.

CHAPTER 4

Implementation

Core Security, Hashing, Protocols, & System Execution

Engineered System Security

🔓 Comprehensive Security Schemes applied across software nodes:

- **JWT Authentication:** Secure token-based session handling.
- **RBAC Middleware:** Dynamic route blocking based on user scope patterns.
- **OWASP Mitigation:** Handled XSS, SQL injections, and CSRF.
- **Sanitization:** Backend input data validation schemas using robust libraries.
- **Data Protection:** Bcrypt password hashing and HTTPS encryption.

CHAPTER 5

Testing

Unit & Integration Test Metrics, Load Testing, & Stability Analysis

Testing Methodology & Summary

Metric Dimension	Target Scope	Executed	Passed	Failed Cases	Coverage Ratio
Unit Testing	Backend Core Services & APIs	15 Cases	✓14 Passed	1 Case (resolved)	**100%**
Integration Testing	Middleware & Database Operations	10 Cases	✓9 Passed	1 Case (resolved)	**100%**
System & Security	RBAC, Forms & CSRF Blocks	5 Cases	✓3 Passed	2 Cases (resolved)	**100%**
Combined Summary	**Overall System Execution**	**30 Cases**	**26 Passed**	**4 Cases**	**100%**

Performance & Load Testing Results

SYSTEM PERFORMANCE BENCHMARK

875/SEC

AVG RESPONSE TIME

JMeter Load Assessment

We stress-tested the server environment mimicking 100 to 500 concurrent users:

- **Max Response Time Under Spike:** 900 ms (Worst Case Scenario)
- **Error Rate:** Less than 2% under peak workloads
- **Robust Stability:** Graceful socket connection degradation without database locking or server crashes.

CHAPTER 6

Results & Evaluation

Business Outcomes, Engineering Milestones, & Deliverables

Key Achievements & Output

Business Outcomes

Successfully delivered all target goals:

- Enabled automated Alumni-University connectivity.
- Secured document processing ecosystem.
- Live real-time communication framework running flawlessly.

Engineering Milestones

Adhered to premium codebase standards:

- Highly scalable, modular RESTful APIs.
- Full deployment-ready DB normalization.
- Robust RBAC middleware protecting private routes.

CHAPTER 7

Future Work

Project Roadmap, AI Integration, & Global Platforms

Roadmap & Future Extensions

AI Moderation

Deploy custom AI models on the backend to automatically review and filter offensive posts and image/video uploads.

Enhanced Media

Integrate robust peer-to-peer video streaming capabilities and custom media-sharing feeds for alumni.

Native App

Develop React Native mobile applications for native push alerting, iOS/Android integration, and QR code widgets.

Thank You

Any Questions?

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